

Computational Ecology & Data Science Transition Syllabus

May 14, 2026 – September 30, 2026

Overview

This syllabus is designed around several constraints and goals:

- Roughly half of working days are field days devoted to bee sampling.
- The goal is not to become an expert programmer immediately.
- The goal is to become computationally fluent and quantitatively confident.
- The emphasis is on transferable skills:
 - Python
 - R
 - GIS/spatial analysis
 - Git/GitHub
 - data wrangling
 - visualization
 - reproducibility
 - ecological analytics
- Every week includes:
 - concepts to focus on
 - practical tasks
 - a small portfolio-style project

The overarching strategy is:

1. Learn concepts.
2. Immediately apply them to real ecological data.
3. Build reusable workflows.
4. Gradually construct a computational portfolio.

General Weekly Structure

Field Days

Focus primarily on:

- data quality
- organization
- consistent metadata collection

- ecological observation
- lightweight field notes

Non-Field Days

Aim for:

- 2–4 focused computational hours/day
- one primary learning objective
- one practical workflow/project

Standing Habits (All Summer)

These continue every week:

1. Maintain a GitHub Repository

Create repositories for:

- bee analyses
- bird analyses
- GIS experiments
- mini-projects

Commit regularly.

2. Keep a Computational Journal

Document:

- commands
- bugs
- solutions
- workflow notes
- analysis ideas
- plotting ideas

3. Keep All Data Tidy

Begin practicing:

- clean file naming
- metadata documentation
- reproducible folder structures

PHASE 1 — FOUNDATIONS

May 14 – June 14

Week 1 (May 14 – May 20)

Theme: Set Up Your Computational Environment

Goals

- Install essential software
- Build workflow foundations
- Begin Python fundamentals

Focus Concepts

- Python basics
- Command line familiarity
- File structures
- Git/GitHub basics

Python Crash Course

Read:

- Chapters 1–2

Tasks

Install:

- Python
- VS Code
- Git
- GitHub Desktop (optional)
- RStudio
- QGIS

Create:

- GitHub account
- first repository
- folder structure for bee project

Learn:

- terminal basics
- navigating directories
- creating/running Python scripts

Mini Project

Create a repository called: `urban-biodiversity-workflows`

Add:

- README
 - project goals
 - folder structure
 - first Python script
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Week 2 (May 21 – May 27)

Theme: Core Python Logic

Focus Concepts

- variables
- loops
- lists
- dictionaries
- functions
- conditionals

Python Crash Course

Read:

- Chapters 3–8

Tasks

Practice:

- loops over species lists
- dictionaries for abundance counts
- writing reusable functions

Mini Project

Create a script that:

- imports a list of bee species
- counts occurrences
- outputs richness and abundance summaries

Bonus: Create a simple “life list tracker” for birds.

Week 3 (May 28 – June 3)

Theme: Working With Real Data

Focus Concepts

- CSV handling
- pandas basics
- dataframes
- filtering
- grouping
- summarizing

Python Crash Course

Read:

- Chapter 10

Tasks

Learn:

- pandas
- reading CSV files
- selecting columns
- filtering observations

Use:

- sample ecological datasets
- or your own field data if available

Mini Project

Build: `bee_data_summary.py`

The script should:

- load field data
 - summarize abundance by site
 - summarize richness by site
 - export summary tables
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Week 4 (June 4 – June 10)

Theme: Visualization Foundations

Focus Concepts

- plotting
- histograms
- scatterplots
- barplots
- visualization principles

Python Crash Course

Read:

- Chapters 15–16

Tasks

Learn:

- matplotlib
- basic seaborn awareness (optional only)
- plot customization

Mini Project

Create:

- species richness plots
- abundance plots
- site comparison figures

Export publication-style figures.

Week 5 (June 11 – June 17)

Theme: Strengthening R Skills

Focus Concepts

- tidyverse
- dplyr
- ggplot2
- data wrangling

Tasks

Recreate your Python analyses in R.

Practice:

- pivoting data
- grouped summaries
- joins
- ggplot visualizations

Mini Project

Create an RMarkdown or Quarto document that:

- imports bee data
- cleans data
- produces plots
- outputs a short report

PHASE 2 — ECOLOGICAL ANALYTICS

June 18 – July 15

Week 6 (June 18 – June 24)

Theme: Ecological Statistics Refresher

Focus Concepts

- richness
- Shannon diversity
- Simpson diversity
- rarefaction
- alpha/beta diversity

Tasks

Learn R ecology packages:

- vegan
- iNEXT (optional)

Mini Project

Compute:

- diversity indices
- accumulation curves
- rarefaction plots

for your bee data.

Week 7 (June 25 – July 1)

Theme: Data Cleaning and Reproducibility

Focus Concepts

- tidy data
- metadata
- reproducibility
- workflow organization

Tasks

Create:

- standardized file naming

- metadata sheets
- reproducible directories

Learn:

- .gitignore
- Git commits
- version control workflow

Mini Project

Turn one week of field data into:

- a fully reproducible workflow from raw data to cleaned output.
-

Week 8 (July 2 – July 8)

Theme: GIS Foundations

Focus Concepts

- coordinate systems
- shapefiles
- vector data
- raster basics
- geospatial thinking

Tasks

Learn:

- QGIS basics
- loading spatial layers
- adding sampling points
- measuring distances

Mini Project

Create a map of:

- all garden sites
- surrounding green space
- borough context

Export publication-quality maps.

Week 9 (July 9 – July 15)

Theme: Spatial Data in R/Python

Focus Concepts

- sf package
- geopandas
- spatial joins
- buffers
- proximity analysis

Tasks

Learn:

- importing shapefiles
- generating buffers
- calculating landscape metrics

Mini Project

Calculate:

- percent green space
- canopy cover
- impervious surface

within multiple buffer distances around gardens.

PHASE 3 — CONNECTIVITY & REMOTE SENSING

July 16 – August 15

Week 10 (July 16 – July 22)

Theme: Remote Sensing Basics

Focus Concepts

- NDVI

- land cover
- canopy cover
- urban heat island
- raster data

Tasks

Explore publicly available datasets.

Learn:

- raster visualization
- clipping rasters
- extracting values

Mini Project

Generate maps showing:

- vegetation density
- canopy cover
- land cover around your sites.

Week 11 (July 23 – July 29)

Theme: Connectivity Thinking

Focus Concepts

- habitat fragmentation
- corridors
- graph theory basics
- resistance surfaces

Tasks

Read introductory papers on:

- landscape connectivity
- urban ecology connectivity

Mini Project

Create a conceptual connectivity model:

- identify likely corridors

- identify isolated gardens
- create connectivity hypotheses

This can initially be exploratory.

Week 12 (July 30 – August 5)

Theme: Statistical Modeling

Focus Concepts

- linear models
- generalized linear models
- model interpretation
- predictors vs response variables

Tasks

Use:

- environmental variables
- landscape variables
- garden traits

as predictors of bee diversity.

Mini Project

Build your first ecological model predicting:

- richness or
- abundance

from environmental variables.

Week 13 (August 6 – August 12)

Theme: Data Visualization & Communication

Focus Concepts

- visual storytelling

- figure design
- dashboard thinking
- communication

Tasks

Improve:

- plots
- labels
- aesthetics
- readability

Mini Project

Create:

- a short presentation or
- a mini dashboard

summarizing preliminary findings.

Week 14 (August 13 – August 19)

Theme: SQL Foundations

Focus Concepts

- relational databases
- SELECT
- JOIN
- GROUP BY
- filtering

Tasks

Use:

- SQLite or
- PostgreSQL basics

Mini Project

Create a database containing:

- bee observations
- garden metadata
- environmental variables

Query it using SQL.

PHASE 4 — ADVANCED INTEGRATION

August 20 – September 30

Week 15 (August 20 – August 26)

Theme: Multi-Taxa Thinking

Focus Concepts

- cross-taxa comparisons
- biodiversity metrics
- urban ecological communities

Tasks

If collecting bird data:

- organize point count data
- compare bee vs bird richness

Mini Project

Create:

- a comparative biodiversity analysis between bees and birds.
-

Week 16 (August 27 – September 2)

Theme: Automation & Workflow Efficiency

Focus Concepts

- scripting repetitive tasks
- reproducible pipelines
- automation mindset

Tasks

Automate:

- imports
- cleaning
- summaries
- exports

Mini Project

Create a script that:

- automatically processes new field data and generates updated summaries/plots.
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Week 17 (September 3 – September 9)

Theme: Intro to Machine Learning Concepts

Focus Concepts

- training/testing
- overfitting
- classification vs regression
- random forests (conceptually)

Tasks

Use scikit-learn for:

- simple predictive analyses

Mini Project

Predict:

- bee richness category or
- diversity level

using environmental variables.

Focus on understanding workflow rather than performance.

Week 18 (September 10 – September 16)

Theme: Scientific Communication Portfolio

Focus Concepts

- portfolio building
- public communication
- documentation

Tasks

Clean and organize:

- GitHub repositories
- READMEs
- notebooks
- plots

Mini Project

Create:

- a polished GitHub repository showcasing one complete workflow.
-

Week 19 (September 17 – September 23)

Theme: Synthesis

Focus Concepts

- integrating datasets
- systems thinking
- dissertation direction refinement

Tasks

Reflect on:

- which analyses excite you most
- which methods feel natural
- where your strengths are emerging

Mini Project

Draft:

- a computationally focused dissertation outline

Potential themes:

- urban biodiversity
 - spatial ecology
 - connectivity
 - socioecological systems
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Week 20 (September 24 – September 30)

Theme: Positioning Your Future Self

Focus Concepts

- career positioning
- transferable skill framing
- long-term planning

Tasks

Update:

- CV
- GitHub
- LinkedIn

Frame yourself as:

- quantitative ecologist
- spatial analyst
- computational urban ecologist
- environmental data scientist

Final Project

Assemble a:

Summer Computational Portfolio

Include:

- maps
- scripts
- models
- figures
- dashboards
- GitHub repositories
- workflow documents

Recommended Long-Term Learning Resources

Python

- Python Crash Course
- Automate the Boring Stuff with Python

R

- R for Data Science
- Geocomputation with R

GIS & Spatial

- QGIS tutorials
- Google Earth Engine tutorials

Statistics

- Statistical Rethinking
 - Applied Predictive Modeling
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The Most Important Goal

By the end of September, success does NOT mean:

- mastering every tool
- becoming an elite programmer
- finishing every analysis

Success means:

- computational thinking feels natural
- coding no longer feels intimidating
- you can independently explore ecological datasets
- you understand reproducible workflows
- you possess visible technical outputs
- your identity has shifted toward quantitative/computational ecology

That shift alone can substantially alter your future trajectory and career optionality.